Syllabus: Flutter for Android Application Development

School: SSCSE		**Batch: 2025-2029**	
Program: B.Tech.		**Current Academic Year: 2025-26**	
Branch: CSE		**Semester: IV**	
1	Course Code	NV62004	Course Name: Flutter for Android Application Development
2	Course Title	Flutter for Android Application Development	
3	Credits	0	
4	Contact Hours (L-T-P)	0-0-2	
	Course Status	Value Added Course	
5	Course Objective	1. Understand the fundamentals of the Dart programming language and object-oriented concepts. 2. Learn the core architecture and design philosophy of Flutter. 3. Build interactive mobile applications using widgets and stateful logic. 4. Apply third-party packages and backend service integration in Flutter. 5. Work with local storage using SQLite and APIs for dynamic data handling. 6. Develop complete mobile applications integrated with Firebase for real-time data and authentication.	
6	Course Outcomes	After the completion of this course students will be able to: CO1: Apply Dart programming fundamentals and implement object-oriented concepts for app logic. CO2: Develop mobile user interfaces using Flutter widgets and layout principles. CO3: Manage application states effectively to create dynamic and responsive user experiences. CO4: Integrate external packages and backend services to build feature-rich applications. CO5: Use SQLite and web APIs for persistent storage and data exchange. CO6: Build and deploy applications with Firebase Database and Authentication features.	
7	Course Description	This course aims to provide students with a comprehensive foundation in mobile application development using both Android and Flutter frameworks. By the end of the course, students will be equipped with the	

		skills to design, develop, and deploy fully functional, data-driven mobile applications across platforms.	
8	Outline syllabus		CO Mapping
	Unit 1	**Introduction to Dart**	
	A	Dart environment setup.	CO1
	B	Basic syntax and control structures.	
	C	OOP concepts in Dart (classes, inheritance).	
	Unit 2	**Basics of Flutter**	
	A	Overview of Flutter framework	CO2
	B	Process of UI building with Widgets	
	C	Stateless vs Stateful widgets utilization	
	Unit 3	**Flutter App State & Packages**	
	A	State management in Flutter using VS code.	CO3, CO4
	B	Introduction to Provider and setState	
	C	Using third-party packages	
	Unit 4	**Local Database and APIs**	
	A	SQLite database integration	CO5
	B	User input via TextFields, Passing data using Navigator	
	C	Overview on REST API integration and location services	
	Unit 5	**Firebase Integration**	
	A	Firebase setup with Flutter	CO6
	B	Cloud Firestore for data storage	
	C	Firebase Authentication (email, Google sign-in) or Oauth authentication.	
	Mode of examination	-	
	Weightage Distribution	ESE	
		100%	
	Text book/s*	1. Beginning Flutter: A Hands-On Guide to App Development – Marco L. Napoli, Wiley	

		2. Flutter for Beginners – Alessandro Biessek, Packt Publishing	
	Other References	Online Resources ● Flutter Official Docs: https://flutter.dev/docs ● Dart Programming Language: https://dart.dev ● Firebase Integration: https://firebase.flutter.dev	

PO and PSO mapping with level of strength for Course:
Table 1

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2	3	-	2	-	-	-	-	2	-	3	-	-
CO2	-	-	3	2	3	-	-	-	-	-	2	3	2	-
CO3	-	2	3		3	-	-	-	-	-	2	2	2	-
CO4	1	1	3	2	2	-	-	-	-	-	2	2	3	-
CO5	-	-	3	2	3	-	-	-	-	-	2	2	2	2
CO6	-	2	3		3	-	-	-	-	-	2	3	3	2

Average of non-zeros entry in following table (should be auto calculated).

Course Code/ Name	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
	Engineering Knowledge	Problem Analysis	Design/ Development of Solutions	Conduct Investigations of Complex Problems	Engineering Tool Usage	The Engineer and The World	Ethics	Individual and Collaborative Team work	Communication	Project Management and Finance	Life-Long Learning	Familiarity and practical proficiency	Understand, analyze and develop	Apply standard Software
Flutter App Dev	1.5	1.4	3.0	1.6	2.8	-	-	-	-	0.7	2.0	2.5	2.3	1.3

Strength of Correlation: 19.1
1. Addressed to ***Slight (Low=1) extent***
2. Addressed to ***Moderate (Medium=2) extent***
3. Addressed to ***Substantial (High=3) extent***